

Practice Worksheet

Section 1: Multiple Choice Questions

(No calculator is allowed for these questions. Select the best answer.)

1. The graph of a periodic function f has a maximum value at $x = 2$ and the adjacent minimum value at $x = 5$. If $f(2) = 10$ and $f(5) = 4$, what is the value of $f(23)$?

- (A) 4
- (B) 7
- (C) 8
- (D) 10

2. A periodic function $g(x)$ is defined for all real numbers. The table below shows some values of $g(x)$. The maximum value of g occurs at $x = 1$, and the adjacent minimum value occurs at $x = 4$. What is the value of $g(16)$?

x	1	4	7
$g(x)$	5	-2	5

- (A) 5
- (B) -2
- (C) 1.5
- (D) 0

3. Let θ be an angle in standard position. If $\cos \theta = -\frac{5}{13}$ and $\tan \theta > 0$, what is the exact value of $\sin \theta$?

- (A) $-\frac{12}{13}$
- (B) $\frac{12}{13}$
- (C) $-\frac{5}{12}$
- (D) $\frac{5}{12}$

4. Evaluate the expression exactly: $\sin\left(\frac{5\pi}{3}\right) \cdot \cos\left(\frac{5\pi}{6}\right)$

- (A) $\frac{3}{4}$

(B) $-\frac{3}{4}$

(C) $\frac{1}{4}$

(D) $-\frac{1}{4}$

5. The point $P(-3, -4)$ lies on the terminal side of angle θ in standard position. What is the value of $\cos \theta - \sin \theta$?

(A) $\frac{1}{5}$

(B) $-\frac{1}{5}$

(C) $\frac{7}{5}$

(D) $-\frac{7}{5}$

6. Let θ be an angle such that $0 < \theta < \frac{\pi}{2}$ and $\sin \theta = k$. Which of the following expressions represents $\cos(\theta + \pi)$ in terms of k ?

(A) $\sqrt{1 - k^2}$

(B) k

(C) $-\sqrt{1 - k^2}$

(D) $-k$

7. Which of the following equations represents a function whose graph has a zero at $(0, 0)$, a minimum at $(\frac{\pi}{2}, -1)$, and returns to zero at $(\pi, 0)$?

(A) $y = \sin x$

(B) $y = -\sin x$

(C) $y = \cos x$

(D) $y = -\cos x$

8. The function $f(x) = \cos x$ is increasing at an *increasing rate* on which of the following intervals?

(A) $(0, \frac{\pi}{2})$

(B) $(\frac{\pi}{2}, \pi)$

(C) $(\pi, \frac{3\pi}{2})$

(D) $(\frac{3\pi}{2}, 2\pi)$

9. What is the phase shift of the sinusoidal function $y = -3 \sin(4x - \pi) + 2$?

(A) Right π (B) Right $\frac{\pi}{4}$ (C) Left π (D) Left $\frac{\pi}{4}$

10. Which of the following functions is equivalent to $g(x) = 5 \sin(2x)$ for all real numbers x ?

(A) $y = 5 \cos\left(2\left(x - \frac{\pi}{4}\right)\right)$

(B) $y = 5 \cos\left(2\left(x + \frac{\pi}{4}\right)\right)$

(C) $y = 5 \cos\left(2\left(x - \frac{\pi}{2}\right)\right)$

(D) $y = -5 \cos(2x)$

11. A Ferris wheel has a radius of 15 meters and its center is 20 meters above the ground. The wheel completes one full revolution every 8 minutes. If a passenger boards the Ferris wheel at its lowest point at $t = 0$ minutes, which of the following functions models the passenger's height $h(t)$ above the ground?

(A) $h(t) = -15 \cos(8t) + 20$

(B) $h(t) = 15 \cos\left(\frac{\pi}{4}t\right) + 20$

(C) $h(t) = -15 \cos\left(\frac{\pi}{4}t\right) + 20$

(D) $h(t) = -15 \cos\left(\frac{\pi}{4}t\right) + 35$

12. The volume of water in a tank fluctuates periodically throughout the day. The table below shows the volume of water, $V(t)$, in gallons, at selected times t , in hours after midnight.

t (hours)	0	6	12	18	24
$V(t)$ (gallons)	40	60	40	20	40

Assuming the function $V(t)$ is sinusoidal and completes exactly one cycle every 24 hours, which of the following equations best models the data?

(A) $V(t) = 20 \cos\left(\frac{\pi}{12}t\right) + 40$

(B) $V(t) = 20 \sin\left(\frac{\pi}{12}t\right) + 40$

(C) $V(t) = 20 \cos\left(\frac{\pi}{6}(t - 6)\right) + 40$

(D) $V(t) = 40 \sin\left(\frac{\pi}{12}t\right) + 20$

Section 2: Free Response Questions

(Show all your work clearly. No calculator is allowed.)

FRQ 1

The depth of water at a pier is modeled by the function $W(t) = 4 \sin\left(\frac{\pi}{6}(t - 3)\right) + 10$, where $W(t)$ is measured in feet and t is the number of hours since midnight ($t = 0$).

- (a) Determine the amplitude, the period, and the midline of the function $W(t)$.
- (b) What is the maximum depth of the water at the pier, and at what time t ($0 \leq t \leq 12$) does this maximum depth first occur?
- (c) Write an equivalent cosine function of the form $W(t) = a \cos(b(t - c)) + d$ that also models the water depth.
- (d) Calculate the average rate of change of the water depth over the interval $0 \leq t \leq 3$. Interpret the meaning of your answer in the context of the problem, including appropriate units.

FRQ 2

Let θ be an angle in standard position such that $\tan \theta = -\frac{8}{15}$ and $\frac{\pi}{2} < \theta < \pi$.

- (a) Determine the exact values of $\sin \theta$ and $\cos \theta$.
- (b) Let α be another angle such that $\alpha = \theta + \pi$. Find the exact value of $\sin \alpha$.
- (c) A point $P(x, y)$ lies on the terminal side of angle θ and is on a circle centered at the origin with a radius of 34. Determine the exact coordinates of P .

FRQ 3

A continuous periodic function $g(x)$ has a minimum value of -2 at $x = 1$ and the adjacent maximum value of 6 at $x = 5$.

- (a) Determine the amplitude, the midline, and the period of $g(x)$.
- (b) Write a formula for $g(x)$ in the form $g(x) = a \cos(b(x - c)) + d$.
- (c) Find all solutions to the equation $g(x) = 2$ on the interval $[1, 9]$.